Supplementary Material 1

Iterative development of the GCS pictograms

The GCS pictograms were developed through an iterative expert-panel evaluation and revision process comprising approximately 20 cycles. During this process, successive versions of the pictograms were reviewed with respect to their comprehensibility, visual clarity, differentiation between GCS response categories, and correspondence with the intended clinical meaning. Expert feedback was used to identify sources of ambiguity and guide subsequent design modifications.

For presentation purposes, the development process is summarized below in five successive sets, each representing approximately four cycles of evaluation and revision. The progression from the initial concepts to the final pictogram set illustrates the principal design problems identified during the expert-panel evaluations and the modifications introduced in response to this feedback.

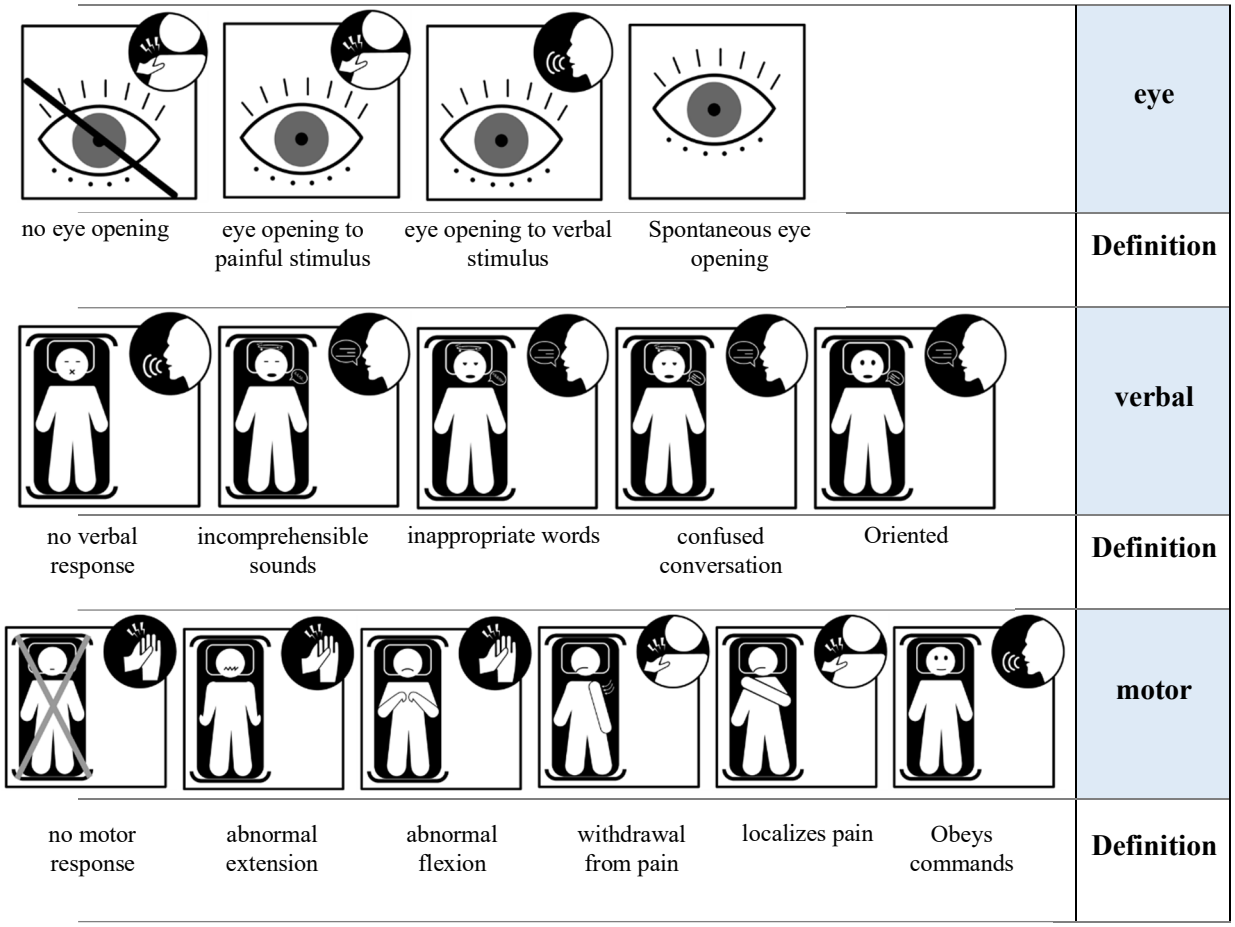
S1.1. First four evaluation–revision cycles

The initial pictograms were designed in black and white using relatively general representations of the human figure. Although the designs incorporated the main concepts underlying the eye, verbal, and motor response categories of the GCS, the relatively high level of visual detail and the use of similar graphical elements across different response categories reduced their visual distinctiveness.

Expert feedback indicated that several pictograms lacked sufficient clarity and that some response categories, particularly those within the verbal-response subscale, were difficult to distinguish. The graphical representation of communication between the nurse and patient was not sufficiently clear to differentiate the verbal-response categories. In addition, inclusion of the patient's bed in the verbal-response pictograms made the verbal and motor subscales visually similar and increased the effort required to distinguish between them.

Based on this feedback, subsequent revisions focused on simplifying the graphical forms, improving the clarity of facial representations, reducing unnecessary visual details, and exploring the use of color to facilitate differentiation between response categories.

Supplementary Fig. S1. Pictogram set after the first four evaluation–revision cycles. These designs did not achieve the predefined level of comprehensibility.



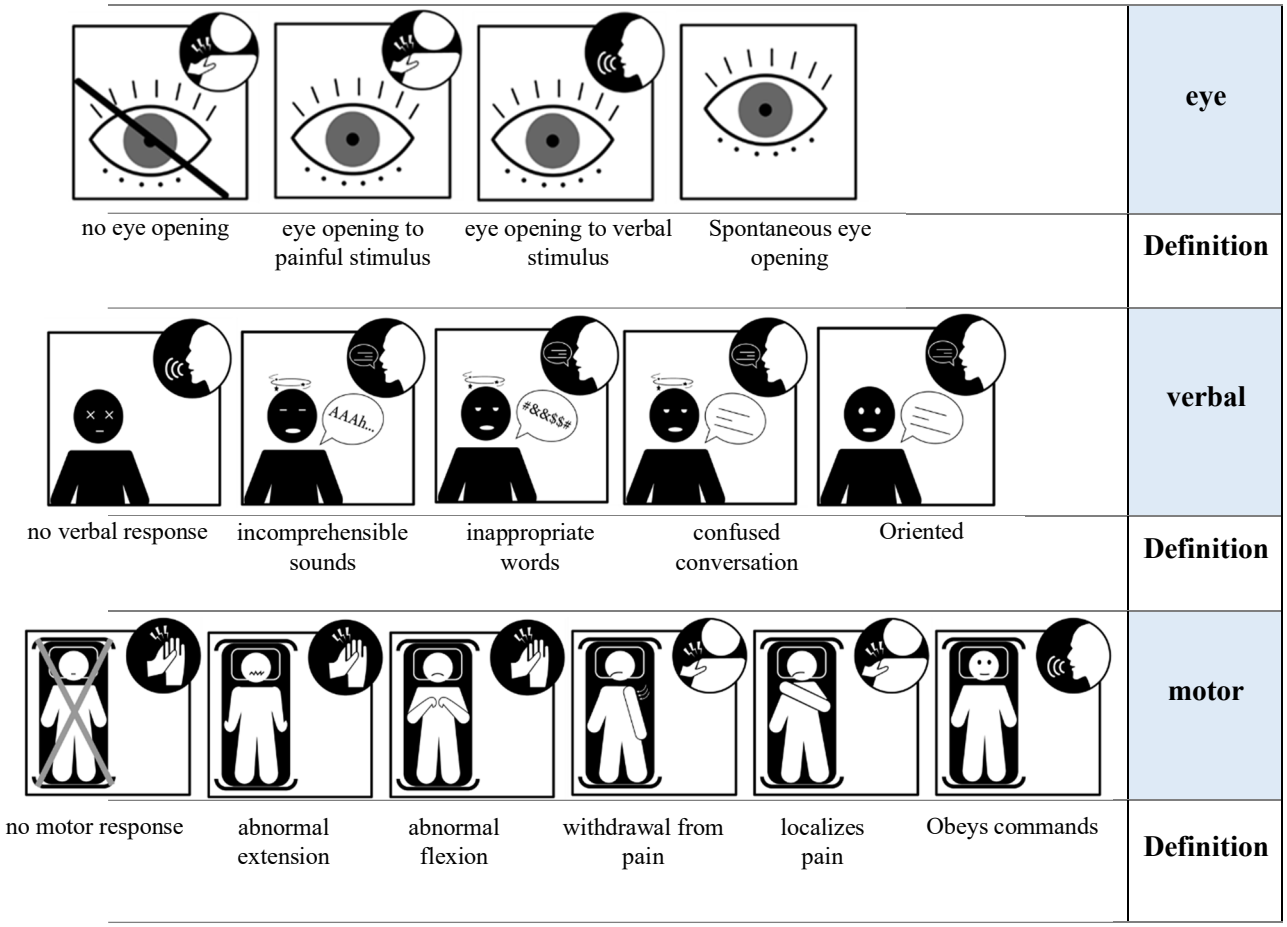
S1.2. Second four evaluation–revision cycles

During the second set of four cycles, particular attention was given to the verbal-response pictograms. The representation of verbal communication became more recognizable following the initial modifications; however, expert feedback indicated that differentiation among individual verbal-response categories remained insufficient. The panel therefore recommended simplifying these pictograms further by focusing primarily on the patient's face and removing contextual elements that did not contribute directly to the intended meaning.

The representation of painful stimulation was also revised. A circular graphical element initially used to indicate a painful stimulus was considered ambiguous and visually complex by some panel members. During intermediate revisions, the textual label “PAIN” was therefore considered as an alternative means of indicating painful stimulation. Different representations of pain application, including stimulation at the shoulder and fingernail, were also reviewed, with the panel recommending greater consistency in the representation of painful stimuli.

Additional changes were made to pictograms that remained difficult to distinguish. The eye-response pictograms were particularly problematic for panel members without nursing or clinical experience, some of whom interpreted the symbols as representing looking or thinking rather than eye opening. These interpretations prompted further modification of the visual cues used to represent eye responses.

Supplementary Fig. S2. Pictogram set after the second four evaluation–revision cycles. Further revision was required because several response categories remained insufficiently differentiated.



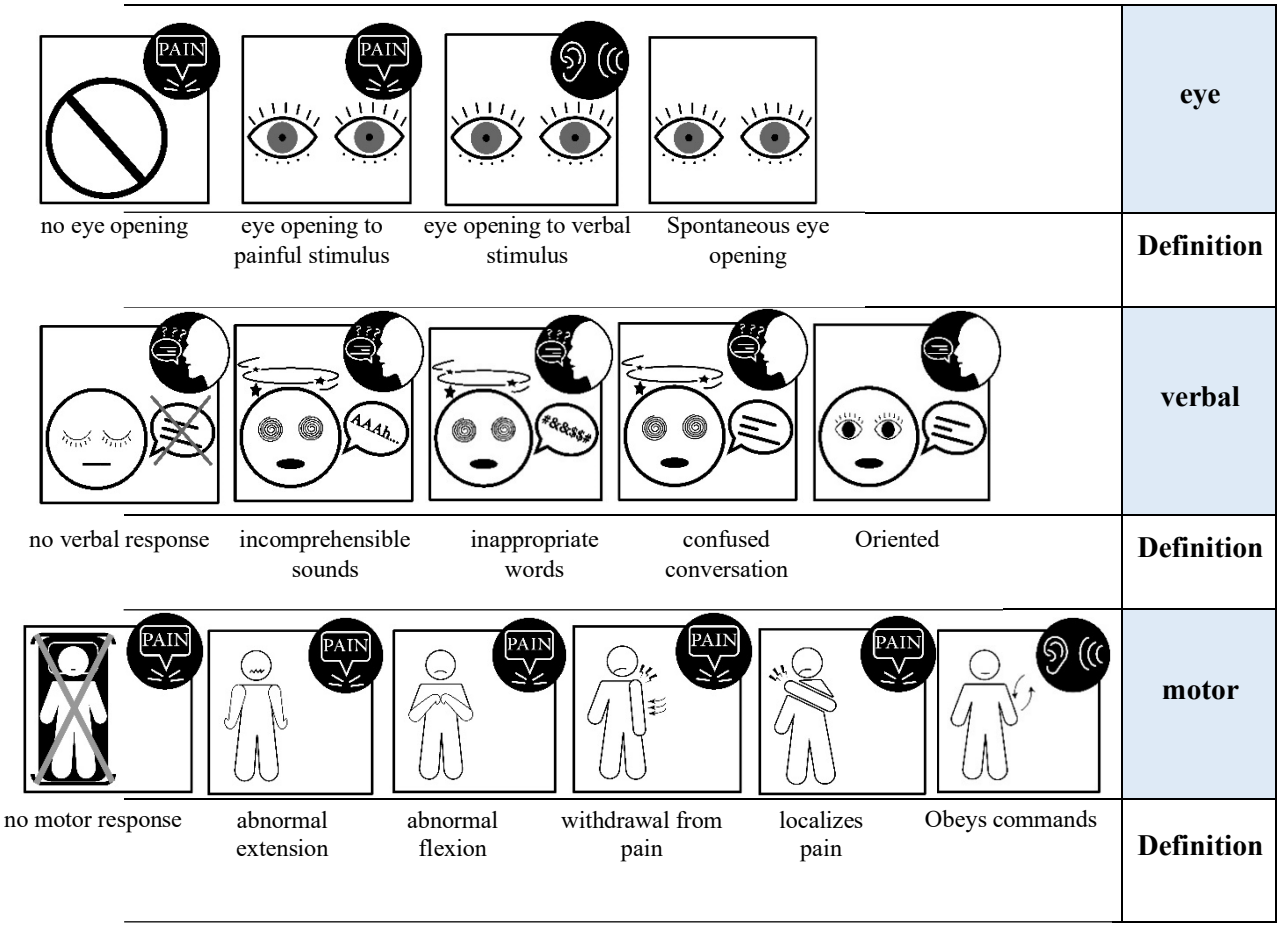
S1.3. Third four evaluation–revision cycles

During the third set of four cycles, additional visual cues were introduced to improve the representation of sensory stimuli and patient responses. Elements such as an ear symbol and indicators of painful stimulation were explored, while the representation of the patient was progressively simplified from a figure including the torso to a predominantly facial representation with more differentiated facial expressions.

Although these modifications improved the perceived representation of verbal communication, expert feedback indicated that some pictograms remained insufficiently distinct from one another. The panel therefore suggested greater use of visual coding and sensory cues associated with the eyes, ears, and mouth to reinforce the intended meanings.

At this stage, the panel also recommended removing unnecessary textual additions, further simplifying the motor-response pictograms, and improving consistency in the representation of non-response categories. The representation of painful stimulation in the motor-response subscale was considered insufficiently clear, and the eye configurations were revised to avoid excessive visual similarity across eye-response categories.

Supplementary Fig. S3. Pictogram set after the third four evaluation–revision cycles. The designs incorporated additional sensory and response-related cues but required further refinement.



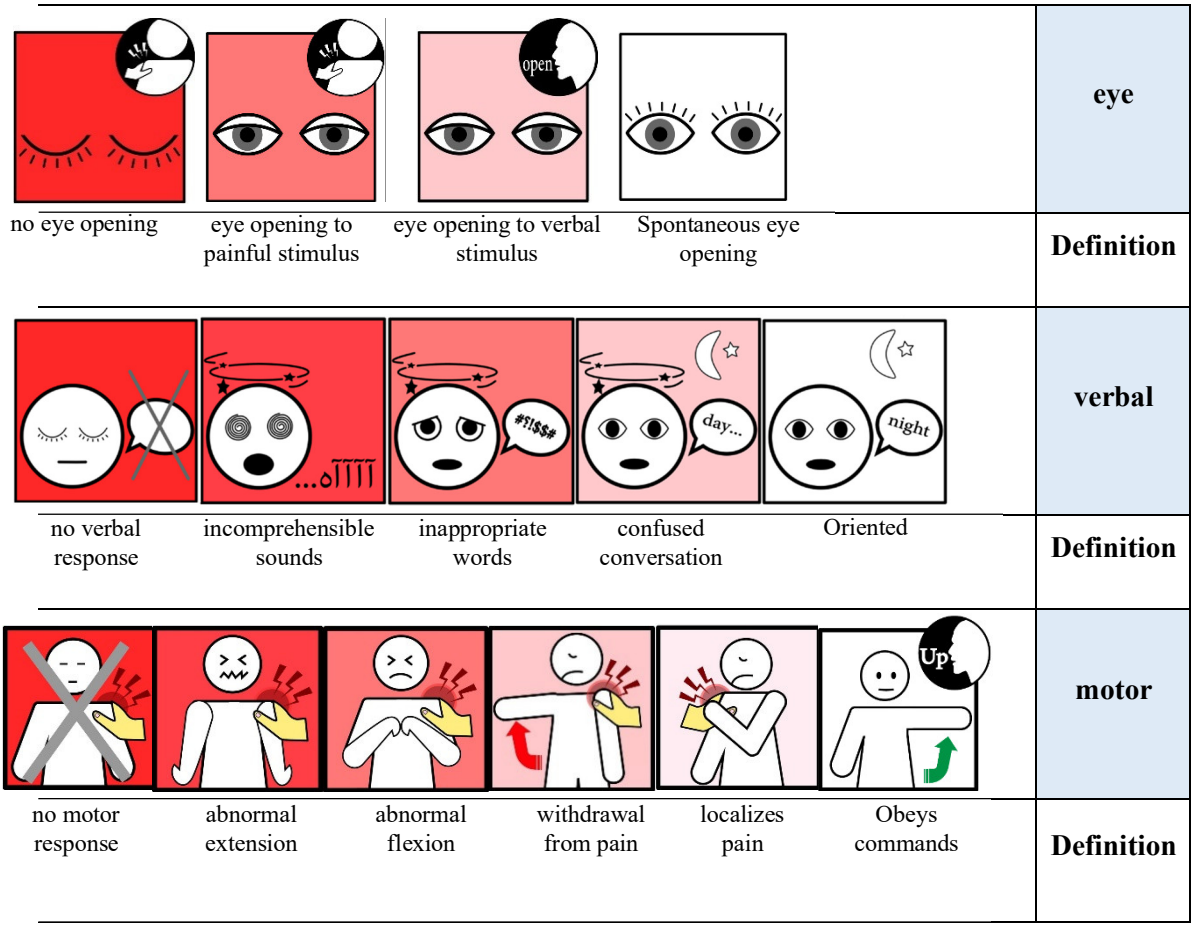
S1.4. Fourth four evaluation–revision cycles

The fourth set of four cycles focused primarily on improving visual consistency across the complete pictogram set. Graphical style, relative element size, background colors, and the arrangement of visual components representing the eyes, ears, and limb movements were adjusted according to the type and level of patient response.

Expert feedback emphasized the need for more expressive and readily interpretable facial features, particularly in the eye- and motor-response pictograms. Panel members also recommended reducing the use of abstract graphical elements, such as representations resembling day/night symbols or halos around the head, because these elements could introduce interpretations unrelated to the intended GCS response.

These revisions shifted the pictogram set toward greater visual consistency, reduced abstraction, and clearer perceptual differentiation between response categories.

Supplementary Fig. S4. Pictogram set after the fourth four evaluation–revision cycles. Although visual consistency had improved, additional modifications were made before finalization.



S1.5. Final four evaluation–revision cycles and selection of the final pictograms

During the final set of evaluation–revision cycles, the designs were further refined to achieve clearer differentiation among the GCS response categories while maintaining a consistent visual language across the complete set. The final pictograms incorporated differentiated color coding, simplified but expressive facial representations, and clearer representations of body and limb movements.

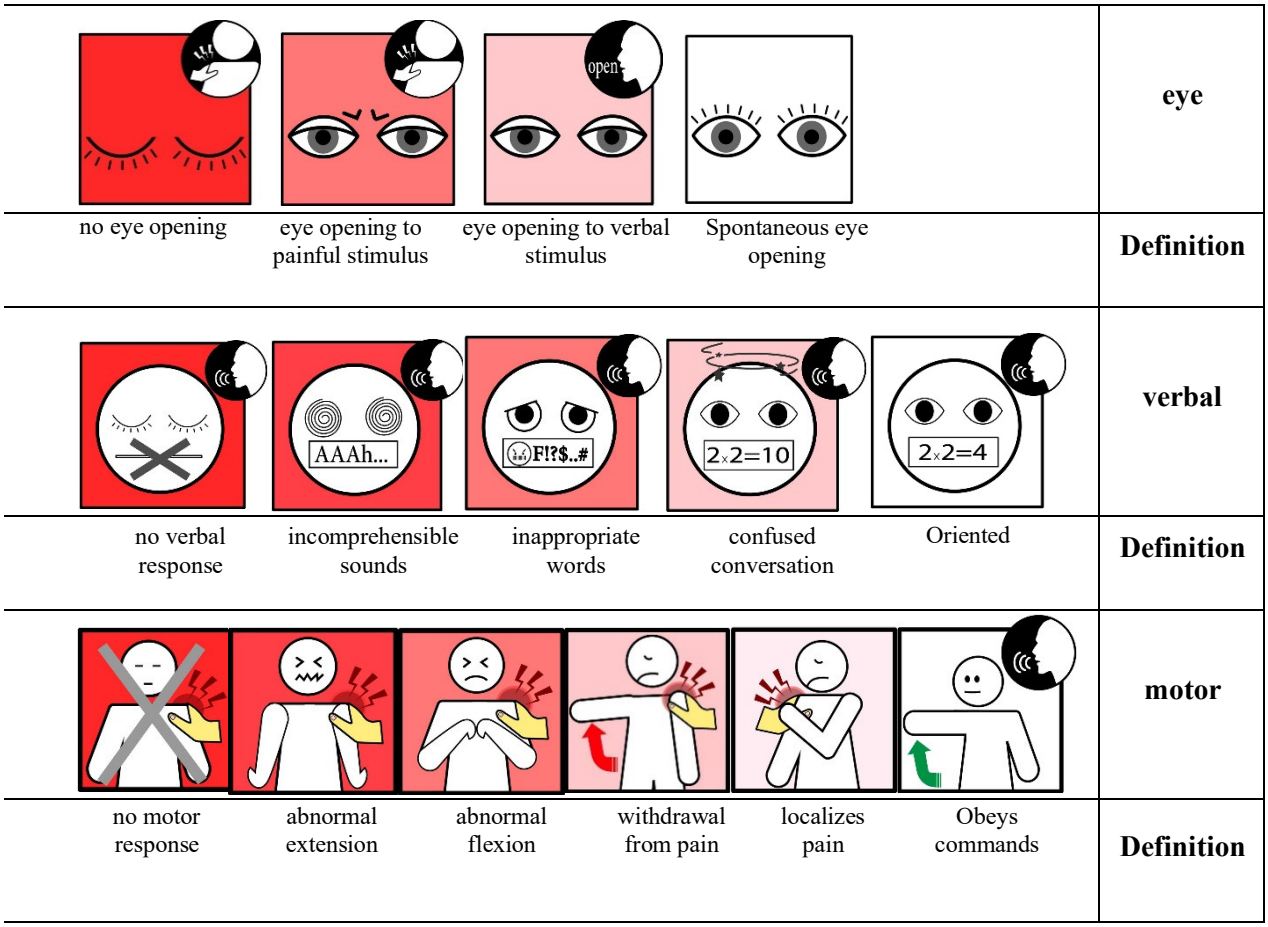
For the eye-response subscale, modifications to eyelid position and related visual cues were used to distinguish spontaneous eye opening, eye opening in response to verbal stimulation, eye opening in response to painful stimulation, and no eye opening.

For the verbal-response subscale, the designs were simplified and differentiated through facial, auditory, and mouth-related visual cues, while unnecessary textual elements introduced during earlier iterations were removed. The final set represented oriented verbal response, confused conversation, inappropriate words, incomprehensible sounds, and no verbal response.

For the motor-response subscale, limb positions and movements were refined to distinguish obeying commands, localizing pain, withdrawal from pain, abnormal flexion, abnormal extension, and no motor response.

Feedback obtained during the final stage indicated improved clarity, simplicity, and correspondence between the pictograms and their intended clinical meanings. Feedback from nursing students during subsequent target-user evaluation was also considered when selecting the final pictogram set.

Supplementary Fig. S5. Final set of GCS pictograms selected following the iterative evaluation–revision process.



Supplementary Material 2

Pictogram Comprehensibility Evaluation (53 participant)

A total of 15 pictograms were developed to represent the response categories of the Glasgow Coma Scale (GCS). Pictogram comprehensibility was evaluated using the ISO 9186-based procedure, and the comprehensibility of each pictogram was compared with the 67% criterion specified for graphical symbols. All 15 pictograms achieved a mean comprehensibility score of at least 67%. The mean and standard deviation of the comprehensibility scores for each pictogram are presented in Table S1.

Table S2. Mean and standard deviation of comprehensibility scores for the 15 GCS pictograms

GCS pictogram	Mean (%)	SD
Eye response (E4)	72.45	43.19
Eye response (E3)	84.41	27.05
Eye response (E2)	86.62	24.61
Eye response (E1)	96.73	12.35
Verbal response (V5)	96.35	13.99
Verbal response (V4)	88.92	18.64
Verbal response (V3)	75.37	26.39
Verbal response (V2)	72.00	26.92
Verbal response (V1)	94.20	21.05
Motor response (M6)	95.90	13.54
Motor response (M5)	68.78	37.94
Motor response (M4)	69.03	33.41
Motor response (M3)	68.89	33.66
Motor response (M2)	69.09	32.32
Motor response (M1)	96.24	15.45

Note. Values are presented as mean percentages and standard deviations. E = eye response; V = verbal response; M = motor response.

Evaluation of the Cognitive Features of the Pictograms

The cognitive characteristics of the pictograms were evaluated across five dimensions: familiarity, simplicity, meaningfulness, semantic closeness, and concreteness. Among these dimensions, familiarity had the lowest overall mean score (48.74), indicating that many of the pictograms were relatively unfamiliar to participants. Meaningfulness (81.94) and semantic closeness (82.25) had the highest overall mean scores, suggesting that the intended concepts were generally well represented by the pictograms. Simplicity (SD = 10.85) and concreteness (SD = 9.25) showed comparatively lower variability across the pictogram set. Overall descriptive statistics for the five cognitive dimensions were: familiarity, 48.74 ± 11.75; simplicity, 72.62 ± 10.85; meaningfulness, 81.94 ± 11.27; semantic closeness, 82.25 ± 11.68; and concreteness, 72.59 ± 9.25.

Taken together, the evaluations indicated relatively strong ratings for meaningfulness and semantic closeness, with moderate-to-high ratings for simplicity and concreteness. Familiarity was comparatively lower, indicating that conceptual correspondence could remain high even when the pictorial representations were not already familiar to users. Scores for each of the 15 pictograms are presented in Table S2.

Table S2. Mean scores for the five cognitive features of each GCS pictogram

GCS pictogram	Familiarity	Simplicity	Meaningfulness	Semantic closeness	Concreteness
Eye response (E4)	70.64	91.44	78.54	89.78	65.24
Eye response (E3)	42.80	82.14	86.23	90.32	77.34
Eye response (E2)	30.90	85.96	87.41	91.60	76.34
Eye response (E1)	34.56	89.46	97.32	95.84	80.25
Verbal response (V5)	68.53	80.20	91.22	87.80	79.36
Verbal response (V4)	56.11	71.16	89.16	86.69	75.47
Verbal response (V3)	49.33	64.87	76.80	76.41	62.66
Verbal response (V2)	39.89	63.92	71.33	72.10	65.24
Verbal response (V1)	55.71	78.99	94.99	93.22	89.79
Motor response (M6)	45.73	65.84	85.11	83.46	86.57
Motor response (M5)	36.14	64.66	67.23	71.77	65.23
Motor response (M4)	40.12	62.32	68.24	70.46	62.48
Motor response (M3)	53.54	63.41	69.78	62.27	64.11
Motor response (M2)	50.64	62.64	67.55	64.35	62.20
Motor response (M1)	56.44	62.30	98.25	97.74	76.58

Note. Scores range from 0 to 100, with higher scores indicating more favorable evaluations of the corresponding cognitive feature. E = eye response; V = verbal response; M = motor response.

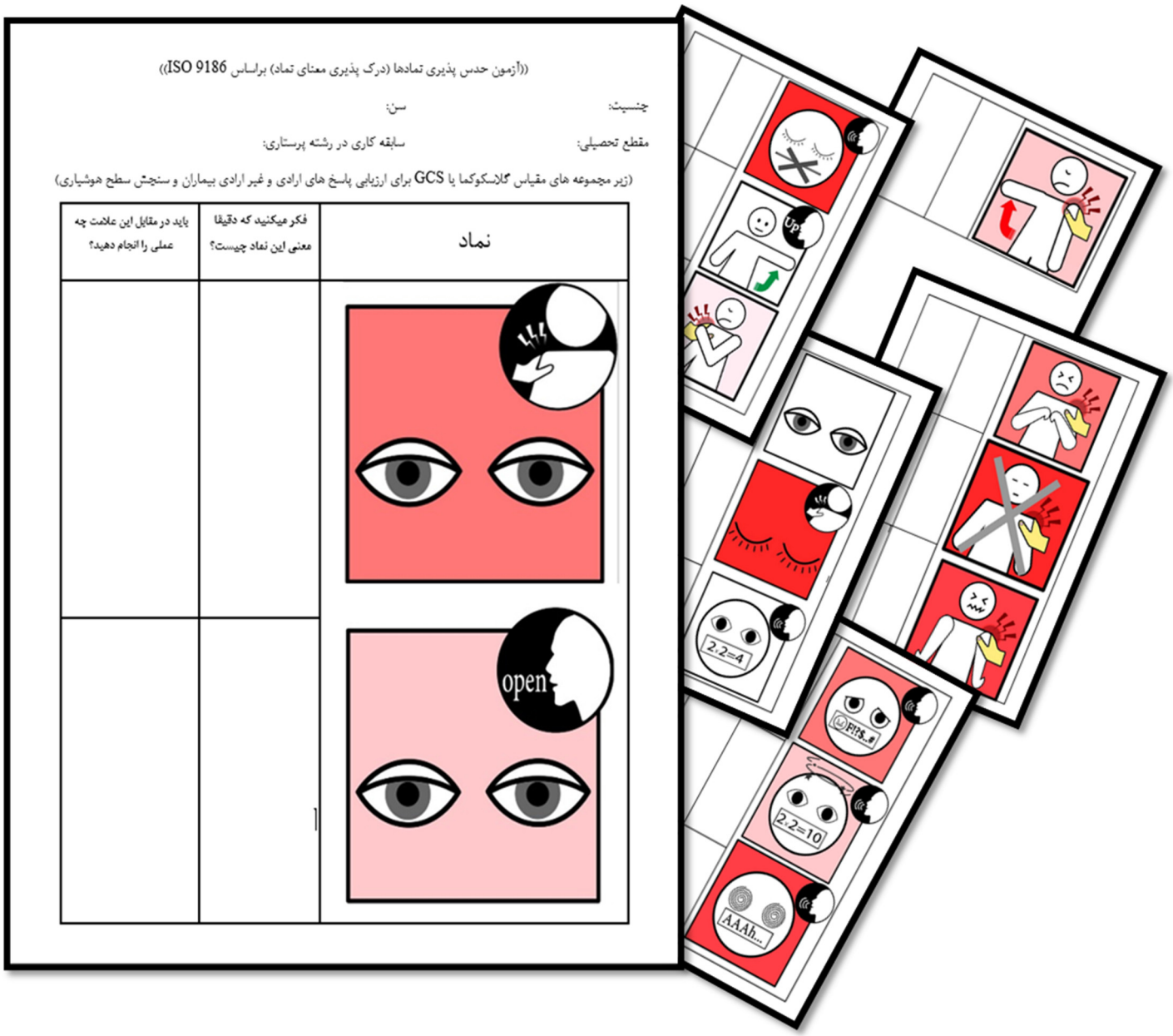
Overall Cognitive Feature Scores Across the 15 GCS Pictograms

Table S3. Descriptive statistics for the five cognitive features across the 15 GCS pictograms

Cognitive feature	n	Minimum	Maximum	Mean	SD
Familiarity	15	30.90	70.64	48.74	11.75
Simplicity	15	62.30	91.44	72.62	10.85
Meaningfulness	15	67.23	98.25	81.94	11.27
Semantic closeness	15	62.27	97.74	82.25	11.68
Concreteness	15	62.20	89.79	72.59	9.25

Note. n represents the number of pictograms included in the summary statistics. Scores range from 0 to 100, with higher scores indicating more favorable evaluations of the corresponding cognitive feature. SD = standard deviation.

Pictogram Comprehensibility Testing Based on ISO 9186



((آزمون حدس پذیری تمادها (درک پذیری معنای تماد) براساس ISO 9186))

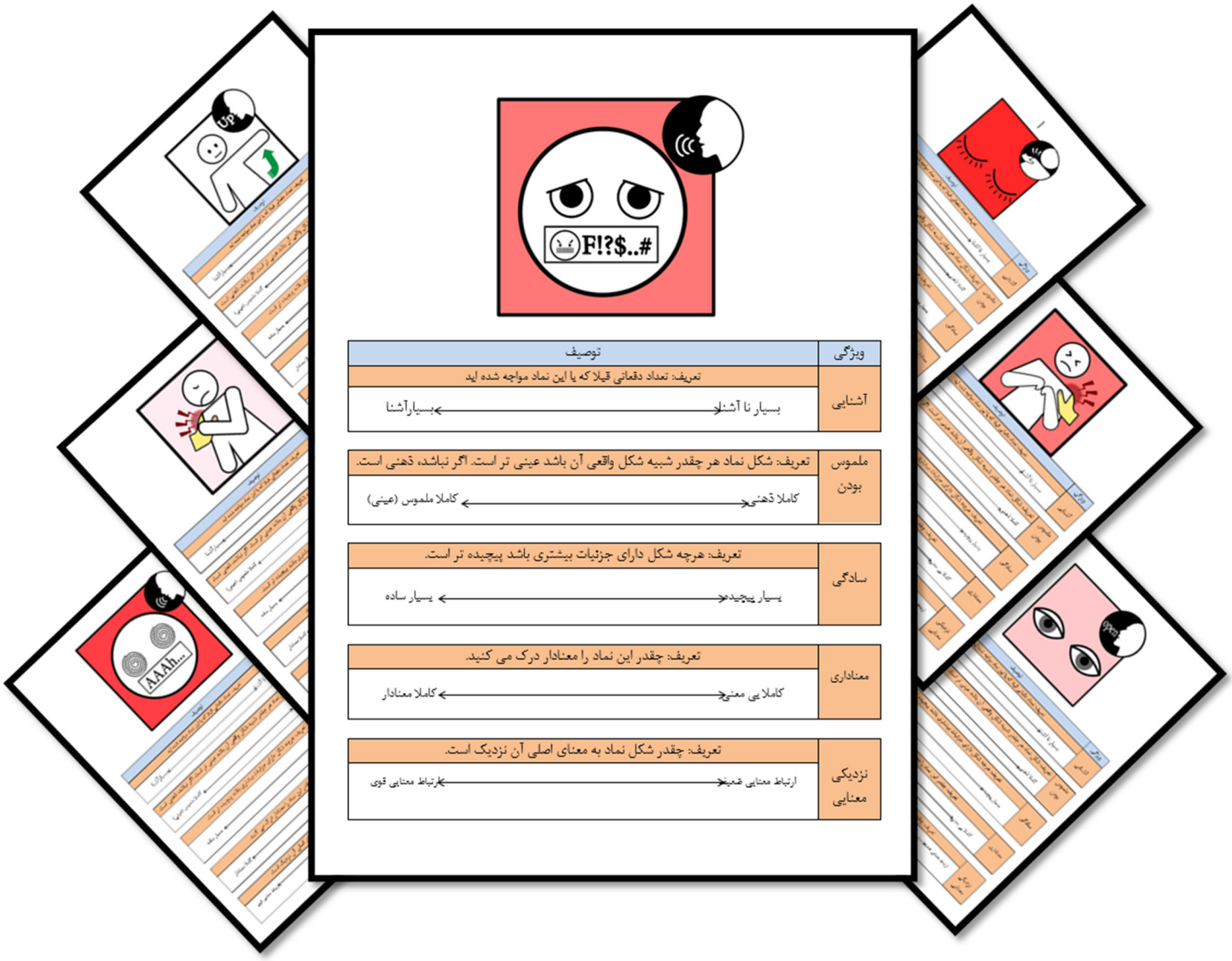
جنسیت: سن:

مقطع تحصیلی: سابقه کاری در رشته پرستاری:

(زیر مجموعه های مقیاس گلاسکوکما یا GCS برای ارزیابی پاسخ های ارادی و غیر ارادی بیماران و سنجش سطح هوشیاری)

نماد	فکر میکنید که دقیقا معنی این تماد چیست؟	باید در مقابل این علامت چه عملی را انجام دهید؟

Cognitive sign Feature Questionnaire



ویژگی	توصیف
آشنایی	تعریف: تعداد دفعاتی قبلا که با این نماد مواجه شده اید بسیار نا آشنا ← → بسیارآشنا
ملموس بودن	تعریف: شکل نماد هر چقدر شبیه شکل واقعی آن باشد عینی تر است. اگر نباشد، ذهنی است. کاملا ذهنی ← → کاملا ملموس (عینی)
سادگی	تعریف: هرچه شکل دارای جزئیات بیشتری باشد پیچیده تر است. بسیار پیچیده ← → بسیار ساده
معناداری	تعریف: چقدر این نماد را معنادار درک می کنید. کاملا بی معنی ← → کاملا معنادار
نزدیکی معنایی	تعریف: چقدر شکل نماد به معنای اصلی آن نزدیک است. ارتباط معنایی ضعیف ← → ارتباط معنایی قوی